DATA-260 HW9 – Caching, Semantic Cache and Self-Improving Agents

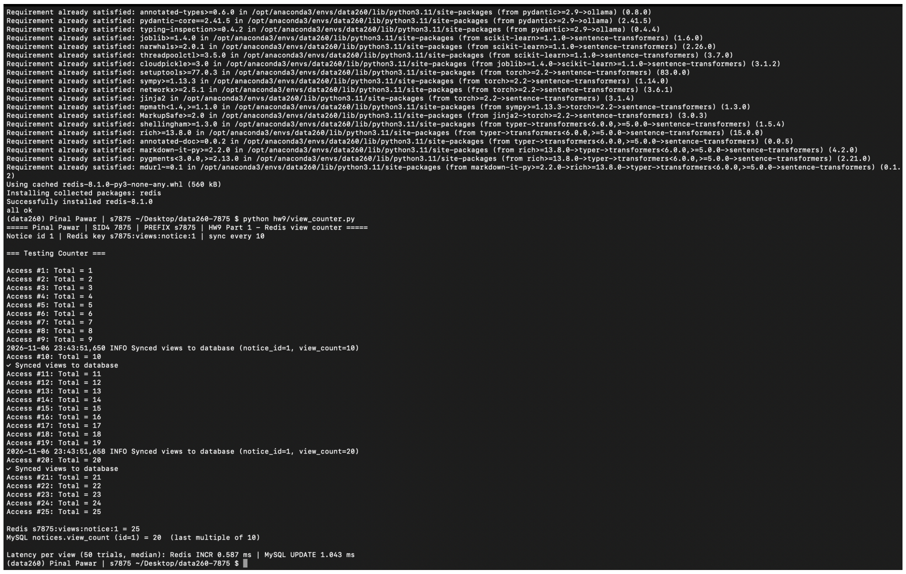
Pinal Pawar | SID4 7875 | PREFIX s7875 | Domain 3: Grocery supply and recall notices
Stack: MySQL `s7875_rel` · Redis Stack 7 (Docker `s7875-redis`) · HW8 Kafka pipeline · qwen3 via Ollama · conda env `data260`
Code: `hw9/` in github.com/PinalPawar/data260-7875

Part 1 – Redis View Counter

`NoticeDBWithCache.increment_notice_views()` (`hw9/view_counter.py`) counts views in Redis and writes them to MySQL only every 10th view (write-behind).

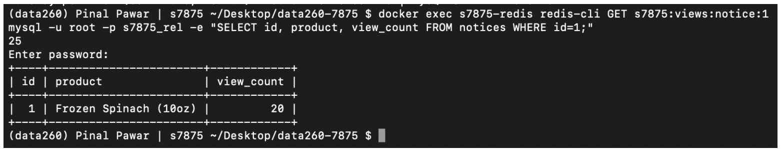
```
def increment_notice_views(self, notice_id: int) -> int:
    count = self.redis_client.incr(f"s7875:views:notice:{notice_id}")  # atomic
    if count % 10 == 0:                                  # persist every 10th view
        conn.execute(text("UPDATE notices SET view_count=:c WHERE id=:id"), ...)
        log.info("Synced views to database (notice_id=%s, count=%s)", notice_id, 
count)
    return count
```

After 25 accesses: Redis = **25**, MySQL = **20** (synced at 10 and 20). Median latency per view (50 trials): Redis INCR **0.59 ms** vs MySQL UPDATE **1.04 ms**. `INCR` is atomic, so concurrent views are never lost. MySQL takes 90% fewer writes; the trade-off is that a Redis crash can lose up to 9 unsynced views.



```
Requirement already satisfied: annotated-types>=0.6.0 in /opt/anaconda3/envs/data260/lib/python3.11/site-packages (from pydantic>=2.9->ollama) (0.8.0)
Requirement already satisfied: pydantic-core==2.41.5 in /opt/anaconda3/envs/data260/lib/python3.11/site-packages (from pydantic>=2.9->ollama) (2.41.5)
Requirement already satisfied: typing-inspection>=0.4.2 in /opt/anaconda3/envs/data260/lib/python3.11/site-packages (from pydantic>=2.9->ollama) (0.4.4)
Requirement already satisfied: joblib>=1.4.0 in /opt/anaconda3/envs/data260/lib/python3.11/site-packages (from scikit-learn>=1.1.0->sentence-transformers) (1.6.0)
Requirement already satisfied: narwhals>=2.0.1 in /opt/anaconda3/envs/data260/lib/python3.11/site-packages (from scikit-learn>=1.1.0->sentence-transformers) (2.26.0)
Requirement already satisfied: threadpoolctl>=3.5.0 in /opt/anaconda3/envs/data260/lib/python3.11/site-packages (from scikit-learn>=1.1.0->sentence-transformers) (3.7.0)
Requirement already satisfied: cloudpickle>=3.0 in /opt/anaconda3/envs/data260/lib/python3.11/site-packages (from joblib>=1.4.0->scikit-learn>=1.1.0->sentence-transformers) (3.1.2)
Requirement already satisfied: setuptools>=77.0.3 in /opt/anaconda3/envs/data260/lib/python3.11/site-packages (from torch>=2.2->sentence-transformers) (83.0.0)
Requirement already satisfied: sympy>=1.13.3 in /opt/anaconda3/envs/data260/lib/python3.11/site-packages (from torch>=2.2->sentence-transformers) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /opt/anaconda3/envs/data260/lib/python3.11/site-packages (from torch>=2.2->sentence-transformers) (3.6.1)
Requirement already satisfied: jinja2 in /opt/anaconda3/envs/data260/lib/python3.11/site-packages (from torch>=2.2->sentence-transformers) (3.1.4)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /opt/anaconda3/envs/data260/lib/python3.11/site-packages (from sympy>=1.13.3->torch>=2.2->sentence-transformers) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /opt/anaconda3/envs/data260/lib/python3.11/site-packages (from jinja2->torch>=2.2->sentence-transformers) (3.0.3)
Requirement already satisfied: shellingham>=1.3.0 in /opt/anaconda3/envs/data260/lib/python3.11/site-packages (from typer->transformers<6.0.0,>=5.0.0->sentence-transformers) (1.5.4)
Requirement already satisfied: rich>=13.8.0 in /opt/anaconda3/envs/data260/lib/python3.11/site-packages (from typer->transformers<6.0.0,>=5.0.0->sentence-transformers) (15.0.0)
Requirement already satisfied: annotated-doc>=0.0.2 in /opt/anaconda3/envs/data260/lib/python3.11/site-packages (from typer->transformers<6.0.0,>=5.0.0->sentence-transformers) (0.0.5)
Requirement already satisfied: markdown-it-py>=2.2.0 in /opt/anaconda3/envs/data260/lib/python3.11/site-packages (from rich>=13.8.0->typer->transformers<6.0.0,>=5.0.0->sentence-transformers) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /opt/anaconda3/envs/data260/lib/python3.11/site-packages (from rich>=13.8.0->typer->transformers<6.0.0,>=5.0.0->sentence-transformers) (2.21.0)
Requirement already satisfied: mdurl~=0.1 in /opt/anaconda3/envs/data260/lib/python3.11/site-packages (from markdown-it-py>=2.2.0->rich>=13.8.0->typer->transformers<6.0.0,>=5.0.0->sentence-transformers) (0.1.
2)
Using cached redis-8.1.0-py3-none-any.whl (560 kB)
Installing collected packages: redis
Successfully installed redis-8.1.0
all ok
(data260) Pinal Pawar | s7875 ~/Desktop/data260-7875 $ python hw9/view_counter.py
===== Pinal Pawar | SID4 7875 | PREFIX s7875 | HW9 Part 1 – Redis view counter =====
Notice id 1 | Redis key s7875:views:notice:1 | sync every 10

=== Testing Counter ===

Access #1: Total = 1
Access #2: Total = 2
Access #3: Total = 3
Access #4: Total = 4
Access #5: Total = 5
Access #6: Total = 6
Access #7: Total = 7
Access #8: Total = 8
Access #9: Total = 9
2026-11-06 23:43:51,650 INFO Synced views to database (notice_id=1, view_count=10)
Access #10: Total = 10
✓ Synced views to database
Access #11: Total = 11
Access #12: Total = 12
Access #13: Total = 13
Access #14: Total = 14
Access #15: Total = 15
Access #16: Total = 16
Access #17: Total = 17
Access #18: Total = 18
Access #19: Total = 19
2026-11-06 23:43:51,658 INFO Synced views to database (notice_id=1, view_count=20)
Access #20: Total = 20
✓ Synced views to database
Access #21: Total = 21
Access #22: Total = 22
Access #23: Total = 23
Access #24: Total = 24
Access #25: Total = 25

Redis s7875:views:notice:1 = 25
MySQL notices.view_count (id=1) = 20  (last multiple of 10)

Latency per view (50 trials, median): Redis INCR 0.587 ms | MySQL UPDATE 1.043 ms
(data260) Pinal Pawar | s7875 ~/Desktop/data260-7875 $ 
```

SS-1: 25 accesses, sync at #10 and #20, latency comparison



```
(data260) Pinal Pawar | s7875 ~/Desktop/data260-7875 $ docker exec s7875-redis redis-cli GET s7875:views:notice:1
mysql -u root -p s7875_rel -e "SELECT id, product, view_count FROM notices WHERE id=1;"
25
Enter password:
+----+----------------------+------------+
| id | product              | view_count |
+----+----------------------+------------+
|  1 | Frozen Spinach (10oz) |         20 |
+----+----------------------+------------+
(data260) Pinal Pawar | s7875 ~/Desktop/data260-7875 $ 
```

SS-2: Redis key = 25, MySQL notices.view_count = 20

Part 2 – Semantic Response Cache

Design

`hw9/semantic_cache.py` sits in front of the HW8 Kafka pipeline (Planner → Writer → Reviewer).

- **Lookup:** each query is embedded with `all-MiniLM-L6-v2` (384-dim). A Redis HNSW cosine index (`s7875:semcache:idx`) returns the nearest cached query with one KNN-1 search; similarity = 1 − cosine distance.
- **Hit (≥ 0.85):** returns the stored answer with `is_cached=true` and the similarity.
- **Miss:** adds the matching `s7875_rel` notice facts, sends the question to `s7875_inbox`, and waits for `s7875_final`. It stores query, embedding, answer, `notice_id` and `data_version`, and returns `is_cached=false`.
- **Logging:** every request logs CACHE HIT/MISS with similarity and latency.
- **Rule:** only Reviewer-approved answers are cached.

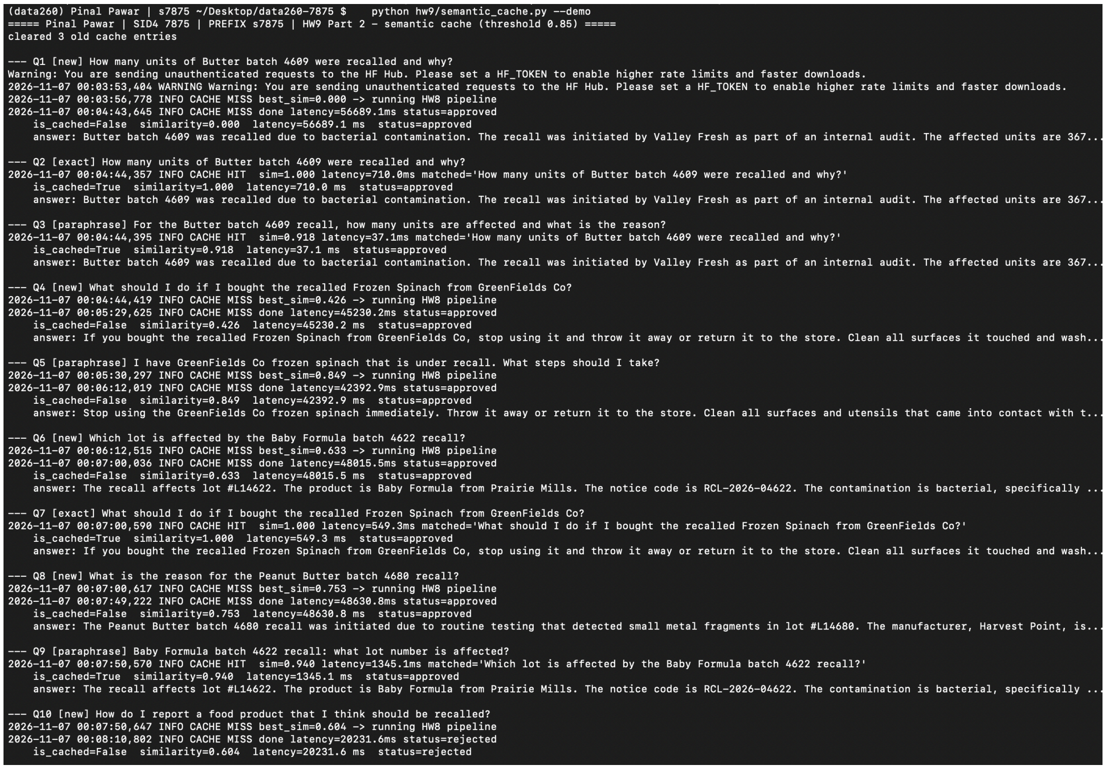
```
def ask(self, query):                                      # hw9/semantic_cache.py (shortened)
    t0 = time.perf_counter(); vec = embed(query)            # all-MiniLM-L6-v2, normalized
    entry, sim = self.nearest(vec, nid)                     # FT.SEARCH ... KNN 1 ... DIALECT 2
    if entry and sim >= self.threshold:                     # HIT: no agent call
        log.info("CACHE HIT sim=%.3f latency=%.1fms", sim, ms(t0))
        return {"answer": entry["answer"], "is_cached": True, "similarity": sim, ...}
    log.info("CACHE MISS best_sim=%.3f -> running HW8 pipeline", sim)
    res = run_pipeline(query + context_line(notice))        # s7875_inbox -> ... -> s7875_final
    if res["status"] == "approved":
        self.store(query, vec, res["answer"], notice_id, data_version(notice), ...)
    return {"answer": res["answer"], "is_cached": False, "similarity": sim, ...}
```

Results (12 queries, threshold 0.85, empty cache)

Type	Queries	Hits	Similarity of hits
new	5	0	–
exact duplicate	3	3	1.000
paraphrase	4	2	0.918, 0.940 (misses: 0.849, 0.572)

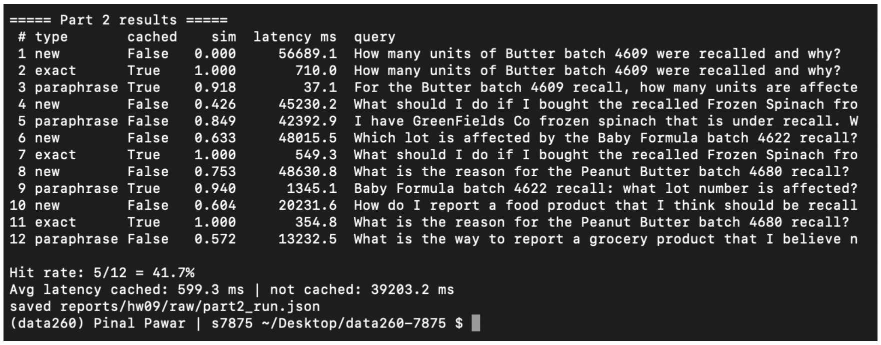
Hit rate 5/12 = 41.7%. Average latency: cached **0.60 s** vs not cached **39.2 s** (about 65× faster).

- **Q5 near-miss:** a true paraphrase scored 0.849, just under the threshold (see Part 3).
- **Q10 and Q12 never cached:** the HW8 regex rule check rejected them as a false positive. The Writer quoted the rule itself ("Never suggest the product might still be safe").
- **Reviewer model:** in a first run, the qwen3:1.7b Reviewer rejected 4 of 9 correct answers with self-contradicting reasons, so even exact duplicates missed. The Reviewer was switched to **qwen3:8b** (`HW8_MODEL=qwen3:8b`); a cache that stores only approved answers is only as good as its approver.



```
(data260) Pinal Pawar | s7875 ~/Desktop/data260-7875 $    python hw9/semantic_cache.py --demo
===== Pinal Pawar | SID4 7875 | PREFIX s7875 | HW9 Part 2 - semantic cache (threshold 0.85) =====
cleared 3 old cache entries

--- Q1 [new] How many units of Butter batch 4609 were recalled and why?
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
2026-11-07 00:03:53,404 WARNING Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
2026-11-07 00:03:56,778 INFO CACHE MISS best_sim=0.000 -> running HW8 pipeline
2026-11-07 00:04:43,645 INFO CACHE MISS done latency=56689.1ms status=approved
    is_cached=False  similarity=0.000  latency=56689.1 ms  status=approved
    answer: Butter batch 4609 was recalled due to bacterial contamination. The recall was initiated by Valley Fresh as part of an internal audit. The affected units are 367...

--- Q2 [exact] How many units of Butter batch 4609 were recalled and why?
2026-11-07 00:04:44,357 INFO CACHE HIT  sim=1.000 latency=710.0ms matched='How many units of Butter batch 4609 were recalled and why?'
    is_cached=True  similarity=1.000  latency=710.0 ms  status=approved
    answer: Butter batch 4609 was recalled due to bacterial contamination. The recall was initiated by Valley Fresh as part of an internal audit. The affected units are 367...

--- Q3 [paraphrase] For the Butter batch 4609 recall, how many units are affected and what is the reason?
2026-11-07 00:04:44,395 INFO CACHE HIT  sim=0.918 latency=37.1ms matched='How many units of Butter batch 4609 were recalled and why?'
    is_cached=True  similarity=0.918  latency=37.1 ms  status=approved
    answer: Butter batch 4609 was recalled due to bacterial contamination. The recall was initiated by Valley Fresh as part of an internal audit. The affected units are 367...

--- Q4 [new] What should I do if I bought the recalled Frozen Spinach from GreenFields Co?
2026-11-07 00:04:44,419 INFO CACHE MISS best_sim=0.426 -> running HW8 pipeline
2026-11-07 00:05:29,625 INFO CACHE MISS done latency=45230.2ms status=approved
    is_cached=False  similarity=0.426  latency=45230.2 ms  status=approved
    answer: If you bought the recalled Frozen Spinach from GreenFields Co, stop using it and throw it away or return it to the store. Clean all surfaces it touched and wash...

--- Q5 [paraphrase] I have GreenFields Co frozen spinach that is under recall. What steps should I take?
2026-11-07 00:05:30,297 INFO CACHE MISS best_sim=0.849 -> running HW8 pipeline
2026-11-07 00:06:12,019 INFO CACHE MISS done latency=42392.9ms status=approved
    is_cached=False  similarity=0.849  latency=42392.9 ms  status=approved
    answer: Stop using the GreenFields Co frozen spinach immediately. Throw it away or return it to the store. Clean all surfaces and utensils that came into contact with t...

--- Q6 [new] Which lot is affected by the Baby Formula batch 4622 recall?
2026-11-07 00:06:12,515 INFO CACHE MISS best_sim=0.633 -> running HW8 pipeline
2026-11-07 00:07:00,036 INFO CACHE MISS done latency=48015.5ms status=approved
    is_cached=False  similarity=0.633  latency=48015.5 ms  status=approved
    answer: The recall affects lot #L14622. The product is Baby Formula from Prairie Mills. The notice code is RCL-2026-04622. The contamination is bacterial, specifically ...

--- Q7 [exact] What should I do if I bought the recalled Frozen Spinach from GreenFields Co?
2026-11-07 00:07:00,590 INFO CACHE HIT  sim=1.000 latency=549.3ms matched='What should I do if I bought the recalled Frozen Spinach from GreenFields Co?'
    is_cached=True  similarity=1.000  latency=549.3 ms  status=approved
    answer: If you bought the recalled Frozen Spinach from GreenFields Co, stop using it and throw it away or return it to the store. Clean all surfaces it touched and wash...

--- Q8 [new] What is the reason for the Peanut Butter batch 4680 recall?
2026-11-07 00:07:00,617 INFO CACHE MISS best_sim=0.753 -> running HW8 pipeline
2026-11-07 00:07:49,222 INFO CACHE MISS done latency=48630.8ms status=approved
    is_cached=False  similarity=0.753  latency=48630.8 ms  status=approved
    answer: The Peanut Butter batch 4680 recall was initiated due to routine testing that detected small metal fragments in lot #L14680. The manufacturer, Harvest Point, is...

--- Q9 [paraphrase] Baby Formula batch 4622 recall: what lot number is affected?
2026-11-07 00:07:50,570 INFO CACHE HIT  sim=0.940 latency=1345.1ms matched='Which lot is affected by the Baby Formula batch 4622 recall?'
    is_cached=True  similarity=0.940  latency=1345.1 ms  status=approved
    answer: The recall affects lot #L14622. The product is Baby Formula from Prairie Mills. The notice code is RCL-2026-04622. The contamination is bacterial, specifically ...

--- Q10 [new] How do I report a food product that I think should be recalled?
2026-11-07 00:07:50,647 INFO CACHE MISS best_sim=0.604 -> running HW8 pipeline
2026-11-07 00:08:10,802 INFO CACHE MISS done latency=20231.6ms status=rejected
    is_cached=False  similarity=0.604  latency=20231.6 ms  status=rejected
```

SS-3: Q1 MISS through the HW8 pipeline, Q2 exact HIT (1.000), Q3 paraphrase HIT (0.918)



```
===== Part 2 results =====
 # type       cached    sim  latency ms  query
 1 new        False   0.000     56689.1  How many units of Butter batch 4609 were recalled and why?
 2 exact      True    1.000       710.0  How many units of Butter batch 4609 were recalled and why?
 3 paraphrase True    0.918        37.1  For the Butter batch 4609 recall, how many units are affecte
 4 new        False   0.426     45230.2  What should I do if I bought the recalled Frozen Spinach fro
 5 paraphrase False   0.849     42392.9  I have GreenFields Co frozen spinach that is under recall. W
 6 new        False   0.633     48015.5  Which lot is affected by the Baby Formula batch 4622 recall?
 7 exact      True    1.000       549.3  What should I do if I bought the recalled Frozen Spinach fro
 8 new        False   0.753     48630.8  What is the reason for the Peanut Butter batch 4680 recall?
 9 paraphrase True    0.940      1345.1  Baby Formula batch 4622 recall: what lot number is affected?
10 new        False   0.604     20231.6  How do I report a food product that I think should be recall
11 exact      True    1.000       354.8  What is the reason for the Peanut Butter batch 4680 recall?
12 paraphrase False   0.572     13232.5  What is the way to report a grocery product that I believe n

Hit rate: 5/12 = 41.7%
Avg latency cached: 599.3 ms | not cached: 39203.2 ms
saved reports/hw09/raw/part2_run.json
(data260) Pinal Pawar | s7875 ~/Desktop/data260-7875 $ 
```

SS-4: Part 2 results table, hit rate and average latency

Part 3 – Threshold Economics and Stale Answers

3.1 Labeled set and method

`hw9/cache_cases.yaml` has 10 seed questions and 40 labeled cases:

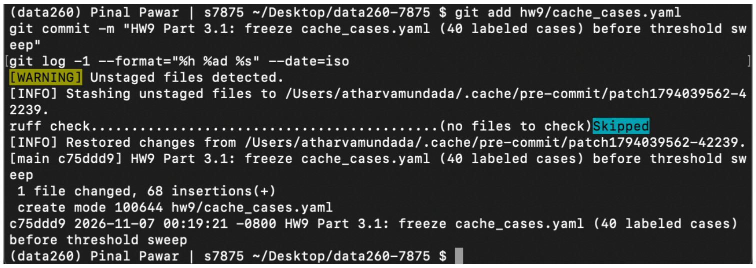
- 10 exact duplicates and 10 paraphrases (expect a hit on the labeled seed);

- 10 similar-but-different (same wording, different batch, different correct answer: expect a miss);
- 10 unrelated (expect a miss).

Freeze: the file was committed before the sweep (`c75ddd9`, 2026-11-07 00:19:21), and the sweep refuses to run if its SHA-256 changes.

Method:

- The seeds were answered once by the HW8 pipeline into a separate index (`s7875:sweep:idx`).
- Each case was searched against that read-only index, so every threshold sees the same cache.
- Every case that could miss ran through the pipeline once (median 38.9 s).
- **Scoring:** a *false hit* is any hit not on the labeled seed. Latency is lookup time on a hit, and lookup plus pipeline time on a miss.



```
(data260) Pinal Pawar | s7875 ~/Desktop/data260-7875 $ git add hw9/cache_cases.yaml
git commit -m "HW9 Part 3.1: freeze cache_cases.yaml (40 labeled cases) before threshold sw
eep"
git log -1 --format="%h %ad %s" --date=iso
[WARNING] Unstaged files detected.
[INFO] Stashing unstaged files to /Users/atharvamundada/.cache/pre-commit/patch1794039562-4
2239.
ruff check...........................................(no files to check)Skipped
[INFO] Restored changes from /Users/atharvamundada/.cache/pre-commit/patch1794039562-42239.
[main c75ddd9] HW9 Part 3.1: freeze cache_cases.yaml (40 labeled cases) before threshold sw
eep
 1 file changed, 68 insertions(+)
 create mode 100644 hw9/cache_cases.yaml
c75ddd9 2026-11-07 00:19:21 -0800 HW9 Part 3.1: freeze cache_cases.yaml (40 labeled cases)
before threshold sweep
(data260) Pinal Pawar | s7875 ~/Desktop/data260-7875 $ 
```

SS-6: cache_cases.yaml committed before the sweep

3.2 Threshold sweep (cosine threshold only)

Threshold	Hit rate	False-hit rate (false/hits)	Mean latency (ms)	Total cost
0.75	75.0%	33.3% (10/30)	7,112	$50.20
0.80	75.0%	33.3% (10/30)	7,112	$50.20
0.85	67.5%	37.0% (10/27)	10,205	$50.26
0.90	67.5%	37.0% (10/27)	10,205	$50.26
0.95	57.5%	39.1% (9/23)	14,424	$45.34

No threshold works. The similar-but-different cases scored **0.910–0.997**, higher than the true paraphrases (0.823–0.969). Raising the threshold drops good paraphrase hits but keeps 9 of the 10 wrong ones. Unrelated questions never exceeded 0.566.

3.3 False hit

D7 “Is the Deli Turkey batch **4624** recall because of Listeria?” matched seed S7 (batch **4619**) at **0.997**.

- **Cached answer:** “…batch 4619 recall is due to **Listeria contamination**…”
- **Correct answer:** “…batch 4624 recall is due to a **mislabeling issue, not Listeria**… an **undeclared allergen**.”

Why it happened: the questions differ only by four digits. Sentence embeddings capture meaning, and a batch number carries almost none, but here the batch number is what decides the answer. An allergic customer would be told the wrong hazard.

Fix (hybrid search): resolve the notice from the question, then run the KNN only over entries tagged with that notice: `(@notice_id:{4624})=>[KNN 1 @embedding $vec]`.

```
pre = f"(@notice_id:{{{notice_id}}})" if notice_id is not None else "*"   # same-
notice filter
res = r.execute_command("FT.SEARCH", index, f"{pre}=>[KNN 1 @embedding $vec AS dist]",
                        "PARAMS", "2", "vec", vec.tobytes(), "SORTBY", "dist", ...,
"DIALECT", "2")
# 3.5 on a hit: re-read the row, compare versions, invalidate if changed
if self.check_version and data_version(get_notice(entry["notice_id"])) !=
entry["data_version"]:
    log.info("CACHE STALE ... -> invalidate %s", entry["key"]);
self.r.delete(entry["key"])
```

3.4 Cost model and chosen threshold

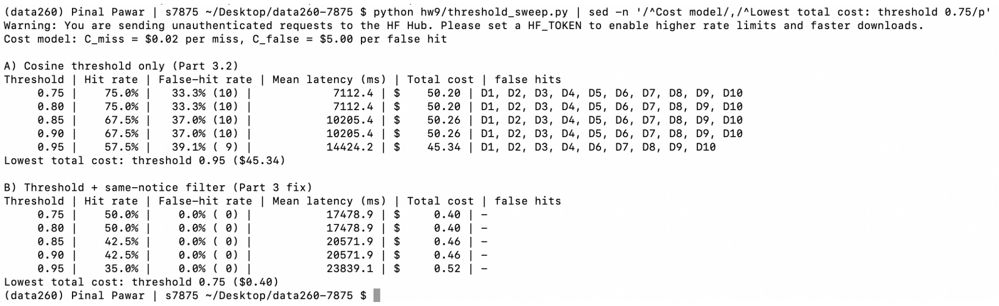
- **C_miss = $0.02:** three LLM calls (about 39 s of GPU time, roughly the price of hosted-API tokens) plus customer wait.
- **C_false = $5.00:** a wrong recall fact triggers at least a support contact (about $5 of agent time) and carries safety and liability risk, so $5 is a conservative floor.
- **Total** = misses × C_miss + false hits × C_false.

Rescored with the same-notice filter (same measurements):

Threshold	Hit rate	False-hit rate	Mean latency (ms)	Total cost
0.75	50.0%	0.0%	17,479	$0.40
0.80	**50.0%**	**0.0%**	**17,479**	**$0.40**
0.85	42.5%	0.0%	20,572	$0.46
0.90	42.5%	0.0%	20,572	$0.46
0.95	35.0%	0.0%	23,839	$0.52

Choice: 0.80 + same-notice filter.

- It has the lowest cost (tied with 0.75): all 20 expected hits are served and there are 0 false hits.
- 0.80 keeps more margin below the weakest paraphrase (0.823).
- **Trade-off:** mean latency rises from 7.1 s to 17.5 s, because the ten look-alike questions now pay for a pipeline run instead of getting a fast wrong answer.



```
(data260) Pinal Pawar | s7875 ~/Desktop/data260-7875 $ python hw9/threshold_sweep.py | sed -n '/^Cost model/,/^Lowest total cost: threshold 0.75/p'
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Cost model: C_miss = $0.02 per miss, C_false = $5.00 per false hit

A) Cosine threshold only (Part 3.2)
Threshold | Hit rate | False-hit rate | Mean latency (ms) | Total cost | false hits
     0.75 |    75.0% |   33.3% (10) |            7112.4 | $    50.20 | D1, D2, D3, D4, D5, D6, D7, D8, D9, D10
     0.80 |    75.0% |   33.3% (10) |            7112.4 | $    50.20 | D1, D2, D3, D4, D5, D6, D7, D8, D9, D10
     0.85 |    67.5% |   37.0% (10) |           10205.4 | $    50.26 | D1, D2, D3, D4, D5, D6, D7, D8, D9, D10
     0.90 |    67.5% |   37.0% (10) |           10205.4 | $    50.26 | D1, D2, D3, D4, D5, D6, D7, D8, D9, D10
     0.95 |    57.5% |   39.1% ( 9) |           14424.2 | $    45.34 | D1, D2, D3, D4, D6, D7, D8, D9, D10
Lowest total cost: threshold 0.95 ($45.34)

B) Threshold + same-notice filter (Part 3 fix)
Threshold | Hit rate | False-hit rate | Mean latency (ms) | Total cost | false hits
     0.75 |    50.0% |    0.0% ( 0) |           17478.9 | $     0.40 | -
     0.80 |    50.0% |    0.0% ( 0) |           17478.9 | $     0.40 | -
     0.85 |    42.5% |    0.0% ( 0) |           20571.9 | $     0.46 | -
     0.90 |    42.5% |    0.0% ( 0) |           20571.9 | $     0.46 | -
     0.95 |    35.0% |    0.0% ( 0) |           23839.1 | $     0.52 | -
Lowest total cost: threshold 0.75 ($0.40)
(data260) Pinal Pawar | s7875 ~/Desktop/data260-7875 $ 
```

SS-7: sweep output – (A) threshold only, (B) with same-notice filter

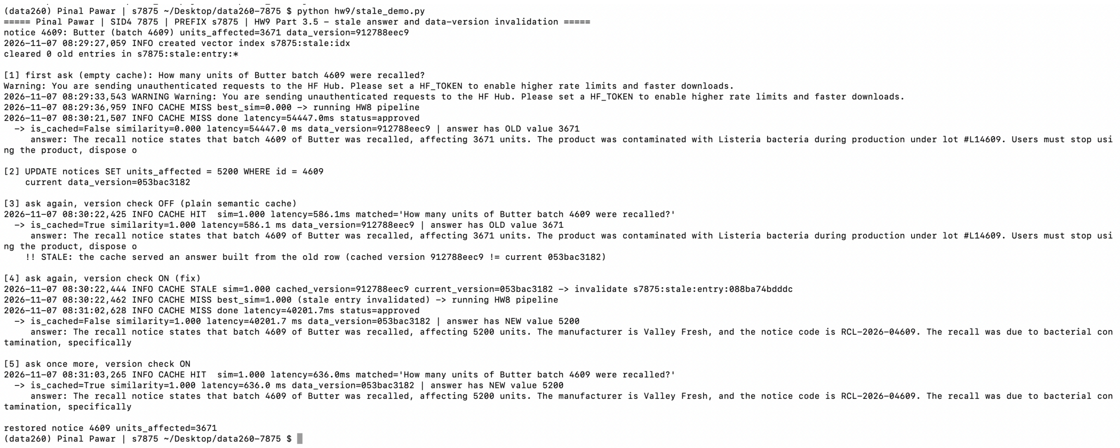
3.5 Stale answer and invalidation

`hw9/stale_demo.py` runs this sequence:

1. **Cache the answer:** “How many units of Butter batch 4609 were recalled?” is cached with **3671** units and `data_version=912788eec9`, a hash of the notice row.
2. **Change the data:** `UPDATE notices SET units_affected=5200`.
3. **Stale hit:** a plain cache HIT at similarity 1.000 still says **3671**.
4. **Version check on:** the cache re-reads the row by primary key on every hit. The versions differ (`053bac3182`), so it logs `CACHE STALE … invalidate`, deletes the entry, and the pipeline answers **5200**.
5. **Fresh hit:** the next ask is a HIT with the new value.

Trade-offs:

- **Version check vs TTL:** a version check never serves an answer built from changed data and never evicts a correct one. It costs one indexed read per hit. A TTL only limits how long staleness lasts.
- **Limit:** it only covers the one row an answer is built from. Multi-row answers need a version per dependency, or invalidation pushed on write (e.g. a Kafka “notice updated” event).



```
(data260) Pinal Pawar | s7875 ~/Desktop/data260-7875 $ python hw9/stale_demo.py
===== Pinal Pawar | SID4 7875 | PREFIX s7875 | HW9 Part 3.5 - stale answer and data-version invalidation =====
notice 4609: Butter (batch 4609) units_affected=3671 data_version=912788eec9
2026-11-07 08:29:27,059 INFO created vector index s7875:stale:idx
cleared 0 old entries in s7875:stale:entry:*

[1] first ask (empty cache): How many units of Butter batch 4609 were recalled?
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
2026-11-07 08:29:33,543 WARNING Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
2026-11-07 08:29:36,959 INFO CACHE MISS best_sim=0.000 -> running HW8 pipeline
2026-11-07 08:30:21,507 INFO CACHE MISS done latency=54447.0ms status=approved
  -> is_cached=False similarity=0.000 latency=54447.0 ms data_version=912788eec9 | answer has OLD value 3671
     answer: The recall notice states that batch 4609 of Butter was recalled, affecting 3671 units. The product was contaminated with Listeria bacteria during production under lot #L14609. Users must stop usi
ng the product, dispose o

[2] UPDATE notices SET units_affected = 5200 WHERE id = 4609
    current data_version=053bac3182

[3] ask again, version check OFF (plain semantic cache)
2026-11-07 08:30:22,425 INFO CACHE HIT  sim=1.000 latency=586.1ms matched='How many units of Butter batch 4609 were recalled?'
  -> is_cached=True similarity=1.000 latency=586.1 ms data_version=912788eec9 | answer has OLD value 3671
     answer: The recall notice states that batch 4609 of Butter was recalled, affecting 3671 units. The product was contaminated with Listeria bacteria during production under lot #L14609. Users must stop usi
ng the product, dispose o
    !! STALE: the cache served an answer built from the old row (cached version 912788eec9 != current 053bac3182)

[4] ask again, version check ON (fix)
2026-11-07 08:30:22,444 INFO CACHE STALE sim=1.000 cached_version=912788eec9 current_version=053bac3182 -> invalidate s7875:stale:entry:088ba74bdddc
2026-11-07 08:30:22,462 INFO CACHE MISS best_sim=1.000 (stale entry invalidated) -> running HW8 pipeline
2026-11-07 08:31:02,628 INFO CACHE MISS done latency=40201.7ms status=approved
  -> is_cached=False similarity=1.000 latency=40201.7 ms data_version=053bac3182 | answer has NEW value 5200
     answer: The recall notice states that batch 4609 of Butter was recalled, affecting 5200 units. The manufacturer is Valley Fresh, and the notice code is RCL-2026-04609. The recall was due to bacterial con
tamination, specifically

[5] ask once more, version check ON
2026-11-07 08:31:03,265 INFO CACHE HIT  sim=1.000 latency=636.0ms matched='How many units of Butter batch 4609 were recalled?'
  -> is_cached=True similarity=1.000 latency=636.0 ms data_version=053bac3182 | answer has NEW value 5200
     answer: The recall notice states that batch 4609 of Butter was recalled, affecting 5200 units. The manufacturer is Valley Fresh, and the notice code is RCL-2026-04609. The recall was due to bacterial con
tamination, specifically

restored notice 4609 units_affected=3671
(data260) Pinal Pawar | s7875 ~/Desktop/data260-7875 $ 
```

SS-9: stale HIT (3671) → CACHE STALE invalidate → re-answer 5200 → fresh HIT

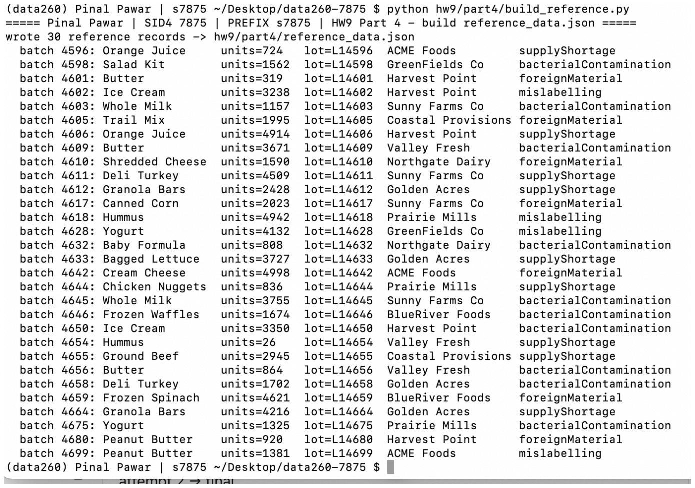
Part 4 – Self-Improving Multi-Agent System

Agents and data

The code is in `hw9/part4/`.

- **Answering Agent:** qwen3:1.7b through the HW8 adapter (temperature 0, fixed seed). It gets the question plus the notice from MySQL, and any relevant lessons.
- **Evaluator Agent:** checks the answer against `reference_data.json`, the ground truth for 30 notices exported from `s7875_rel`. For each question type it checks units, lot, manufacturer, notice code, reason, the required consumer action, no invented recall details, and the HW8 never-say-safe rule.
- **Why rules, not an LLM judge:** Part 2 showed a 1.7b LLM judge rejecting correct answers. The LLM is used only to word new lessons.
- **Datasets:** `training_data.json` (12), `validation_data.json` (10) and `test_data.json` (12) contain questions only, including 2 unrelated questions each.

```
class EvaluatorAgent:
    def evaluate(self, question, answer):
        ref = find_reference(question, self.reference)   # ground-truth record
        # units / lot / manufacturer / code / reason checks by question type ...
        if not any(w in answer.lower() for w in ACTION_WORDS):
            codes.append("MISSING_ACTION")
        return {"result": "PASS" or "REVISE", "explanation": ...,
                "feedback": ..., "failure_codes": codes}

def run_question(question, answerer, evaluator, *, repair, store=None, allowed=None):
    lessons = store.relevant(question, allowed) if store else []  # lessons first
    answer = answerer.answer(question, lessons)
    ev = evaluator.evaluate(question, answer)
    if repair and ev["result"] == "REVISE":                       # max 2 attempts
        answer = answerer.answer(question, lessons, previous=answer,
                                 feedback=ev["feedback"])
        ev = evaluator.evaluate(question, answer)
```



```
(data260) Pinal Pawar | s7875 ~/Desktop/data260-7875 $ python hw9/part4/build_reference.py
===== Pinal Pawar | SID4 7875 | PREFIX s7875 | HW9 Part 4 – build reference_data.json =====
wrote 30 reference records -> hw9/part4/reference_data.json
  batch 4596: Orange Juice     units=724   lot=L14596  ACME Foods         supplyShortage
  batch 4598: Salad Kit        units=1562  lot=L14598  GreenFields Co     bacterialContamination
  batch 4601: Butter           units=319   lot=L14601  Harvest Point      foreignMaterial
  batch 4602: Ice Cream        units=3238  lot=L14602  Harvest Point      mislabelling
  batch 4603: Whole Milk       units=1157  lot=L14603  Sunny Farms Co     bacterialContamination
  batch 4605: Trail Mix        units=1995  lot=L14605  Coastal Provisions foreignMaterial
  batch 4606: Orange Juice     units=4914  lot=L14606  Harvest Point      supplyShortage
  batch 4609: Butter           units=3671  lot=L14609  Valley Fresh       bacterialContamination
  batch 4610: Shredded Cheese  units=1590  lot=L14610  Northgate Dairy    foreignMaterial
  batch 4611: Deli Turkey      units=4509  lot=L14611  Sunny Farms Co     supplyShortage
  batch 4612: Granola Bars     units=2428  lot=L14612  Golden Acres       supplyShortage
  batch 4617: Canned Corn      units=2023  lot=L14617  Sunny Farms Co     foreignMaterial
  batch 4618: Hummus           units=4942  lot=L14618  Prairie Mills      mislabelling
  batch 4628: Yogurt           units=4132  lot=L14628  GreenFields Co     mislabelling
  batch 4632: Baby Formula     units=808   lot=L14632  Northgate Dairy    bacterialContamination
  batch 4633: Bagged Lettuce   units=3727  lot=L14633  Golden Acres       supplyShortage
  batch 4642: Cream Cheese     units=4998  lot=L14642  ACME Foods         foreignMaterial
  batch 4644: Chicken Nuggets  units=836   lot=L14644  Prairie Mills      supplyShortage
  batch 4645: Whole Milk       units=3755  lot=L14645  Sunny Farms Co     bacterialContamination
  batch 4646: Frozen Waffles   units=1674  lot=L14646  BlueRiver Foods    bacterialContamination
  batch 4650: Ice Cream        units=3350  lot=L14650  Harvest Point      bacterialContamination
  batch 4654: Hummus           units=26    lot=L14654  Valley Fresh       supplyShortage
  batch 4655: Ground Beef      units=2945  lot=L14655  Coastal Provisions supplyShortage
  batch 4656: Butter           units=864   lot=L14656  Valley Fresh       bacterialContamination
  batch 4658: Deli Turkey      units=1702  lot=L14658  Golden Acres       bacterialContamination
  batch 4659: Frozen Spinach   units=4621  lot=L14659  BlueRiver Foods    foreignMaterial
  batch 4664: Granola Bars     units=4216  lot=L14664  Golden Acres       supplyShortage
  batch 4675: Yogurt           units=1325  lot=L14675  Prairie Mills      bacterialContamination
  batch 4680: Peanut Butter    units=920   lot=L14680  Harvest Point      foreignMaterial
  batch 4699: Peanut Butter    units=1381  lot=L14699  ACME Foods         mislabelling
(data260) Pinal Pawar | s7875 ~/Desktop/data260-7875 $ 
```

SS-10: reference_data.json built from s7875_rel (30 notices)

4.1 Revision sequence (controlled case)

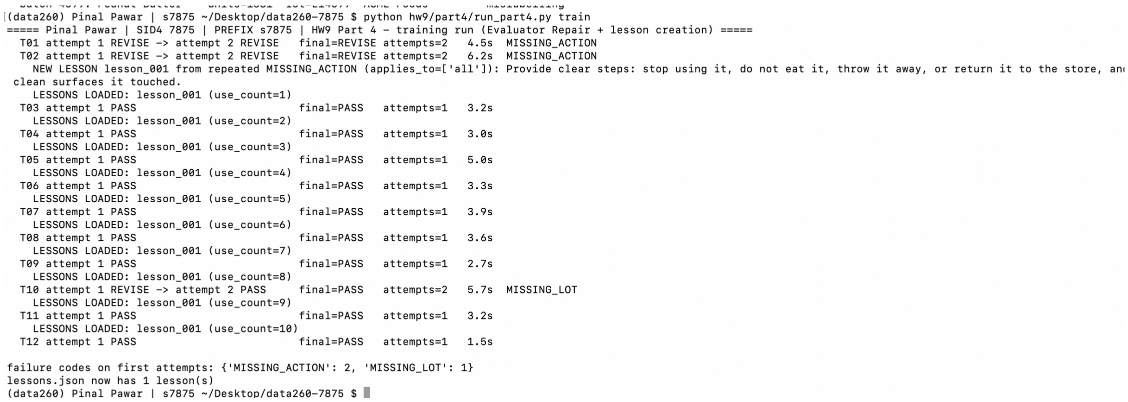
For “How many units of Butter batch 4609 were recalled?” attempt 1 is deliberately given no notice.

- **Attempt 1:** “No recall notice is available for Butter batch 4609.”
- **Evaluator: REVISE**, codes MISSING_UNITS and MISSING_ACTION, with feedback to state the exact units and tell the customer what to do.
- **Attempt 2:** “The recall affects 3,671 units of Butter batch 4609. Stop using it, do not eat it, and throw it away…”
- **Final: PASS** after **2 attempts** (5.5 s).

4.2 Lesson creation and reuse

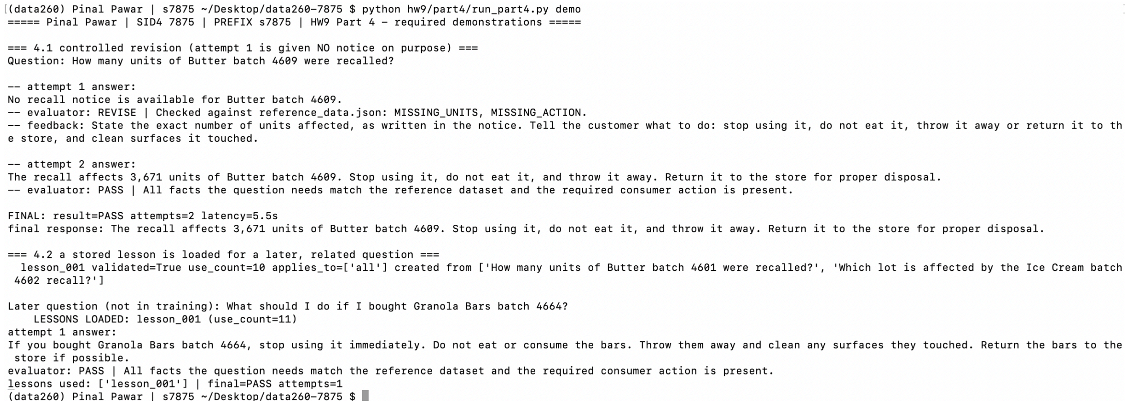
- **Creation:** in training, MISSING_ACTION failed on T01 and T02, so the repeated pattern created `lesson_001` (“Provide clear steps: stop using it, do not eat it, throw it away, or return it to the store, and clean surfaces it touched”, applies to all question types).
- **Reuse in training:** the lesson was loaded for T03–T12, and 9 of the remaining 10 passed on the first attempt.
- **Later question:** “What should I do if I bought Granola Bars batch 4664?” (not in any dataset) loaded `lesson_001` and passed on attempt 1.

```
{"lesson_id": "lesson_001",
 "failure_pattern": "Missing action instructions for the customer",
 "correction": "Provide clear steps: stop using it, do not eat it, throw it away,
                 or return it to the store, and clean surfaces it touched.",
 "created_at": "2026-11-07", "use_count": 35, "applies_to": ["all"], "validated":
true}
```



```
[(data260) Pinal Pawar | s7875 ~/Desktop/data260-7875 $ python hw9/part4/run_part4.py train
===== Pinal Pawar | SID4 7875 | PREFIX s7875 | HW9 Part 4 - training run (Evaluator Repair + lesson creation) =====
  T01 attempt 1 REVISE -> attempt 2 REVISE   final=REVISE attempts=2   4.5s  MISSING_ACTION
  T02 attempt 1 REVISE -> attempt 2 REVISE   final=REVISE attempts=2   6.2s  MISSING_ACTION
    NEW LESSON lesson_001 from repeated MISSING_ACTION (applies_to=['all']): Provide clear steps: stop using it, do not eat it, throw it away, or return it to the store, an
 clean surfaces it touched.
    LESSONS LOADED: lesson_001 (use_count=1)
  T03 attempt 1 PASS                         final=PASS   attempts=1   3.2s
    LESSONS LOADED: lesson_001 (use_count=2)
  T04 attempt 1 PASS                         final=PASS   attempts=1   3.0s
    LESSONS LOADED: lesson_001 (use_count=3)
  T05 attempt 1 PASS                         final=PASS   attempts=1   5.0s
    LESSONS LOADED: lesson_001 (use_count=4)
  T06 attempt 1 PASS                         final=PASS   attempts=1   3.3s
    LESSONS LOADED: lesson_001 (use_count=5)
  T07 attempt 1 PASS                         final=PASS   attempts=1   3.9s
    LESSONS LOADED: lesson_001 (use_count=6)
  T08 attempt 1 PASS                         final=PASS   attempts=1   3.6s
    LESSONS LOADED: lesson_001 (use_count=7)
  T09 attempt 1 PASS                         final=PASS   attempts=1   2.7s
    LESSONS LOADED: lesson_001 (use_count=8)
  T10 attempt 1 REVISE -> attempt 2 PASS     final=PASS   attempts=2   5.7s  MISSING_LOT
    LESSONS LOADED: lesson_001 (use_count=9)
  T11 attempt 1 PASS                         final=PASS   attempts=1   3.2s
    LESSONS LOADED: lesson_001 (use_count=10)
  T12 attempt 1 PASS                         final=PASS   attempts=1   1.5s

failure codes on first attempts: {'MISSING_ACTION': 2, 'MISSING_LOT': 1}
lessons.json now has 1 lesson(s)
(data260) Pinal Pawar | s7875 ~/Desktop/data260-7875 $ █
```

SS-11: training run – lesson_001 created after the repeated MISSING_ACTION failure, then loaded



```
[(data260) Pinal Pawar | s7875 ~/Desktop/data260-7875 $ python hw9/part4/run_part4.py demo
===== Pinal Pawar | SID4 7875 | PREFIX s7875 | HW9 Part 4 - required demonstrations =====

=== 4.1 controlled revision (attempt 1 is given NO notice on purpose) ===
Question: How many units of Butter batch 4609 were recalled?

-- attempt 1 answer:
No recall notice is available for Butter batch 4609.
-- evaluator: REVISE | Checked against reference_data.json: MISSING_UNITS, MISSING_ACTION.
-- feedback: State the exact number of units affected, as written in the notice. Tell the customer what to do: stop using it, do not eat it, throw it away or return it to th
e store, and clean surfaces it touched.

-- attempt 2 answer:
The recall affects 3,671 units of Butter batch 4609. Stop using it, do not eat it, and throw it away. Return it to the store for proper disposal.
-- evaluator: PASS | All facts the question needs match the reference dataset and the required consumer action is present.

FINAL: result=PASS attempts=2 latency=5.5s
final response: The recall affects 3,671 units of Butter batch 4609. Stop using it, do not eat it, and throw it away. Return it to the store for proper disposal.

=== 4.2 a stored lesson is loaded for a later, related question ===
  lesson_001 validated=True use_count=10 applies_to=['all'] created from ['How many units of Butter batch 4601 were recalled?', 'Which lot is affected by the Ice Cream batch
 4602 recall?']

Later question (not in training): What should I do if I bought Granola Bars batch 4664?
    LESSONS LOADED: lesson_001 (use_count=11)
attempt 1 answer:
If you bought Granola Bars batch 4664, stop using it immediately. Do not eat or consume the bars. Throw them away and clean any surfaces they touched. Return the bars to the
 store if possible.
evaluator: PASS | All facts the question needs match the reference dataset and the required consumer action is present.
lessons used: ['lesson_001'] | final=PASS attempts=1
(data260) Pinal Pawar | s7875 ~/Desktop/data260-7875 $ █
```

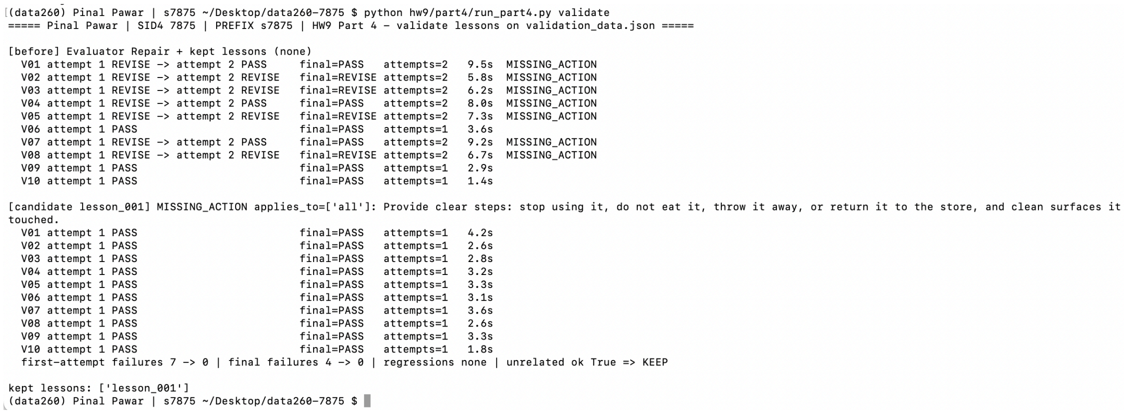
SS-13: 4.1 revision sequence and 4.2 lesson reuse on a later question

4.3 Validation

On `validation_data.json` with Evaluator Repair, with and without `lesson_001`:

- First-attempt failures: **7 → 0**.
- Final failures: **4 → 0**.
- No previously correct answer regressed, and both unrelated questions still passed.

Decision: KEEP. If any condition had failed, the lesson would have been marked REMOVE and left out of the Validated variant. The test set was used only once per variant, after this decision.

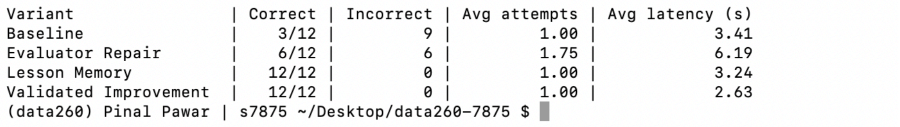


```
(data260) Pinal Pawar | s7875 ~/Desktop/data260-7875 $ python hw9/part4/run_part4.py validate
===== Pinal Pawar | SID4 7875 | PREFIX s7875 | HW9 Part 4 - validate lessons on validation_data.json =====

[before] Evaluator Repair + kept lessons (none)
  V01 attempt 1 REVISE -> attempt 2 PASS     final=PASS   attempts=2   9.5s  MISSING_ACTION
  V02 attempt 1 REVISE -> attempt 2 REVISE   final=REVISE attempts=2   5.8s  MISSING_ACTION
  V03 attempt 1 REVISE -> attempt 2 REVISE   final=REVISE attempts=2   6.2s  MISSING_ACTION
  V04 attempt 1 REVISE -> attempt 2 PASS     final=PASS   attempts=2   8.0s  MISSING_ACTION
  V05 attempt 1 REVISE -> attempt 2 REVISE   final=REVISE attempts=2   7.3s  MISSING_ACTION
  V06 attempt 1 PASS                         final=PASS   attempts=1   3.6s
  V07 attempt 1 REVISE -> attempt 2 PASS     final=PASS   attempts=2   9.2s  MISSING_ACTION
  V08 attempt 1 REVISE -> attempt 2 REVISE   final=REVISE attempts=2   6.7s  MISSING_ACTION
  V09 attempt 1 PASS                         final=PASS   attempts=1   2.9s
  V10 attempt 1 PASS                         final=PASS   attempts=1   1.4s

[candidate lesson_001] MISSING_ACTION applies_to=['all']: Provide clear steps: stop using it, do not eat it, throw it away, or return it to the store, and clean surfaces it
touched.
  V01 attempt 1 PASS                         final=PASS   attempts=1   4.2s
  V02 attempt 1 PASS                         final=PASS   attempts=1   2.6s
  V03 attempt 1 PASS                         final=PASS   attempts=1   2.8s
  V04 attempt 1 PASS                         final=PASS   attempts=1   3.2s
  V05 attempt 1 PASS                         final=PASS   attempts=1   3.3s
  V06 attempt 1 PASS                         final=PASS   attempts=1   3.1s
  V07 attempt 1 PASS                         final=PASS   attempts=1   3.6s
  V08 attempt 1 PASS                         final=PASS   attempts=1   2.6s
  V09 attempt 1 PASS                         final=PASS   attempts=1   3.3s
  V10 attempt 1 PASS                         final=PASS   attempts=1   1.8s
  first-attempt failures 7 -> 0 | final failures 4 -> 0 | regressions none | unrelated ok True => KEEP

kept lessons: ['lesson_001']
(data260) Pinal Pawar | s7875 ~/Desktop/data260-7875 $ 
```

SS-12: validation before/after lesson_001 → KEEP

4.4 Four variants on test_data.json (12 held-out questions)

Variant	Correct answers	Incorrect answers	Average attempts	Average latency
Baseline	3	9	1.00	3.41 s
Evaluator Repair	6	6	1.75	6.19 s
Lesson Memory	12	0	1.00	3.24 s
Validated Improvement	12	0	1.00	2.63 s



```
Variant                | Correct | Incorrect | Avg attempts | Avg latency (s)
Baseline               |    3/12 |         9 |         1.00 |            3.41
Evaluator Repair       |    6/12 |         6 |         1.75 |            6.19
Lesson Memory          |   12/12 |         0 |         1.00 |            3.24
Validated Improvement  |   12/12 |         0 |         1.00 |            2.63
(data260) Pinal Pawar | s7875 ~/Desktop/data260-7875 $ 
```

SS-15: final results table (full per-question output in reports/hw09/raw/part4_test_results.json)

- **Baseline:** mostly failed one way: the answer stated the fact but did not tell the customer what to do.
- **Evaluator Repair:** fixed half of the failures, but cost a second model call (+2.8 s).
- **The lesson:** moved the fix to the first attempt, so it was both correct and fast.
- **Lesson Memory vs Validated:** they are identical here because training produced one lesson and it passed validation. The validation step decided nothing new on this data, but it guards against an over-general lesson in future runs.

Communication, stopping, validation, limitation

- **Communication:** the agents exchange JSON messages. The Answering Agent sends `{question, answer, attempt}`; the Evaluator replies `{result, explanation, feedback, failure_codes}`. Feedback is appended to the revision prompt, and every exchange is kept in a trace.
- **Lesson reuse:** lessons are tagged by question type (or “all”), loaded before the first answer, and `use_count` is incremented.
- **Stopping condition:** PASS, or stop after the 2nd attempt.
- **Validation:** a lesson is kept only if validation failures do not increase, nothing that was correct breaks, and unrelated questions stay clean.
- **Limitation:** the Evaluator checks facts by keyword and number matching. It can pass an answer that contains the right number but wrong reasoning, and it can fail a correct answer that is phrased differently (e.g. “discard” vs “dispose of”). Also, with 12 test questions and a single lesson, these results are a small sample.

Appendix – Datasets

cache_cases.yaml (frozen in commit c75ddd9) – seeds

S1	How many units of Butter batch 4609 were recalled?
S2	What is the reason for the Peanut Butter batch 4680 recall?
S3	Which lot is affected by the Baby Formula batch 4622 recall?
S4	Who manufactures the Yogurt batch 4682 that was recalled?
S5	What should I do if I bought the recalled Frozen Spinach from GreenFields Co?
S6	What category is the Ice Cream batch 4602 recall?
S7	Is the Deli Turkey batch 4619 recall because of Listeria?
S8	How many units of Orange Juice batch 4606 are affected?
S9	What is the notice code for the Hummus batch 4618 recall?
S10	Which company recalled the Granola Bars batch 4612?

cache_cases.yaml – 40 labeled cases (seed = expected match; for misses, the look-alike seed)

ID	Type	Expect	Seed	Query
E1	exact	hit	S1	How many units of Butter batch 4609 were recalled?
E2	exact	hit	S2	What is the reason for the Peanut Butter batch 4680 recall?
E3	exact	hit	S3	Which lot is affected by the Baby Formula batch 4622 recall?
E4	exact	hit	S4	Who manufactures the Yogurt batch 4682 that was recalled?
E5	exact	hit	S5	What should I do if I bought the recalled Frozen Spinach from GreenFields Co?
E6	exact	hit	S6	What category is the Ice Cream batch 4602 recall?
E7	exact	hit	S7	Is the Deli Turkey batch 4619 recall because of Listeria?

ID	Type	Expect	Seed	Query
E8	exact	hit	S8	How many units of Orange Juice batch 4606 are affected?
E9	exact	hit	S9	What is the notice code for the Hummus batch 4618 recall?
E10	exact	hit	S10	Which company recalled the Granola Bars batch 4612?
P1	paraphrase	hit	S1	For Butter batch 4609, what is the number of recalled units?
P2	paraphrase	hit	S2	Why was Peanut Butter batch 4680 recalled?
P3	paraphrase	hit	S3	Baby Formula batch 4622 recall: what lot number does it cover?
P4	paraphrase	hit	S4	Which manufacturer makes the recalled Yogurt batch 4682?

ID	Type	Expect	Seed	Query
P5	paraphrase	hit	S5	I have GreenFields Co frozen spinach that is under recall. What steps should I take?
P6	paraphrase	hit	S6	Ice Cream batch 4602 was recalled - which recall category does it fall under?
P7	paraphrase	hit	S7	Was Deli Turkey batch 4619 recalled due to Listeria contamination ?
P8	paraphrase	hit	S8	What number of Orange Juice batch 4606 units does the recall affect?
P9	paraphrase	hit	S9	Give me the recall notice code for Hummus batch 4618.
P10	paraphrase	hit	S10	Who is the company behind the Granola Bars batch 4612 recall?

ID	Type	Expect	Seed	Query
D1	similar_different	miss	S1	How many units of Butter batch 4656 were recalled?
D2	similar_different	miss	S2	What is the reason for the Peanut Butter batch 4699 recall?
D3	similar_different	miss	S3	Which lot is affected by the Baby Formula batch 4632 recall?
D4	similar_different	miss	S4	Who manufactures the Yogurt batch 4628 that was recalled?
D5	similar_different	miss	S5	What should I do if I bought the recalled Frozen Spinach batch 4626?
D6	similar_different	miss	S6	What category is the Ice Cream batch 4604 recall?
D7	similar_different	miss	S7	Is the Deli Turkey batch 4624 recall because of Listeria?

ID	Type	Expect	Seed	Query
D8	similar_different	miss	S8	How many units of Orange Juice batch 4596 are affected?
D9	similar_different	miss	S9	What is the notice code for the Hummus batch 4654 recall?
D10	similar_different	miss	S10	Which company recalled the Granola Bars batch 4635?
U1	unrelated	miss		What is the capital of France?
U2	unrelated	miss		How do I reset my password on the grocery store website?
U3	unrelated	miss		What time does the store open on Sundays?
U4	unrelated	miss		Can you give me a recipe for banana bread?
U5	unrelated	miss		How long should I boil an egg for a soft yolk?

ID	Type	Expect	Seed	Query
U6	unrelated	miss		What is the difference between Listeria and Salmonella?
U7	unrelated	miss		Can I freeze sliced bread to keep it fresh longer?
U8	unrelated	miss		How do I return an item to the store without a receipt?
U9	unrelated	miss		Do you sell gluten-free pasta?
U10	unrelated	miss		Who won the last FIFA World Cup?

Part 4 input files contain only an id and a question (no expected answers). The ground truth is `hw9/part4/reference_data.json` (30 notices; fields: notice_id, batch, notice_code, product, manufacturer, category, reason, units_affected, lot, description, required_action), shown in SS-10.

training_data.json (12 questions)

T01	How many units of Butter batch 4601 were recalled?
T02	Which lot is affected by the Ice Cream batch 4602 recall?
T03	Who manufactures the recalled Whole Milk batch 4603?
T04	Why was Trail Mix batch 4605 recalled?
T05	What is the notice code for the Orange Juice batch 4606 recall?
T06	Is the Salad Kit batch 4598 recall because of Listeria?
T07	How many units of Granola Bars batch 4612 are affected?
T08	Which company recalled the Hummus batch 4618?
T09	What should I do if I bought Canned Corn batch 4617?
T10	Which lot is affected by the Deli Turkey batch 4611 recall?
T11	What is a good recipe for pancakes?
T12	What are the store's holiday hours?

validation_data.json (10 questions)

V01	How many units of Yogurt batch 4628 were recalled?
V02	Which lot is affected by the Baby Formula batch 4632 recall?
V03	Who manufactures the recalled Cream Cheese batch 4642?
V04	Why was Chicken Nuggets batch 4644 recalled?
V05	Is the Whole Milk batch 4645 recall because of Listeria?
V06	What should I do if I bought Frozen Waffles batch 4646?
V07	What is the notice code for the Ground Beef batch 4655 recall?
V08	Which company recalled the Peanut Butter batch 4680?
V09	How do I cook rice in a microwave?
V10	Do you offer home delivery?

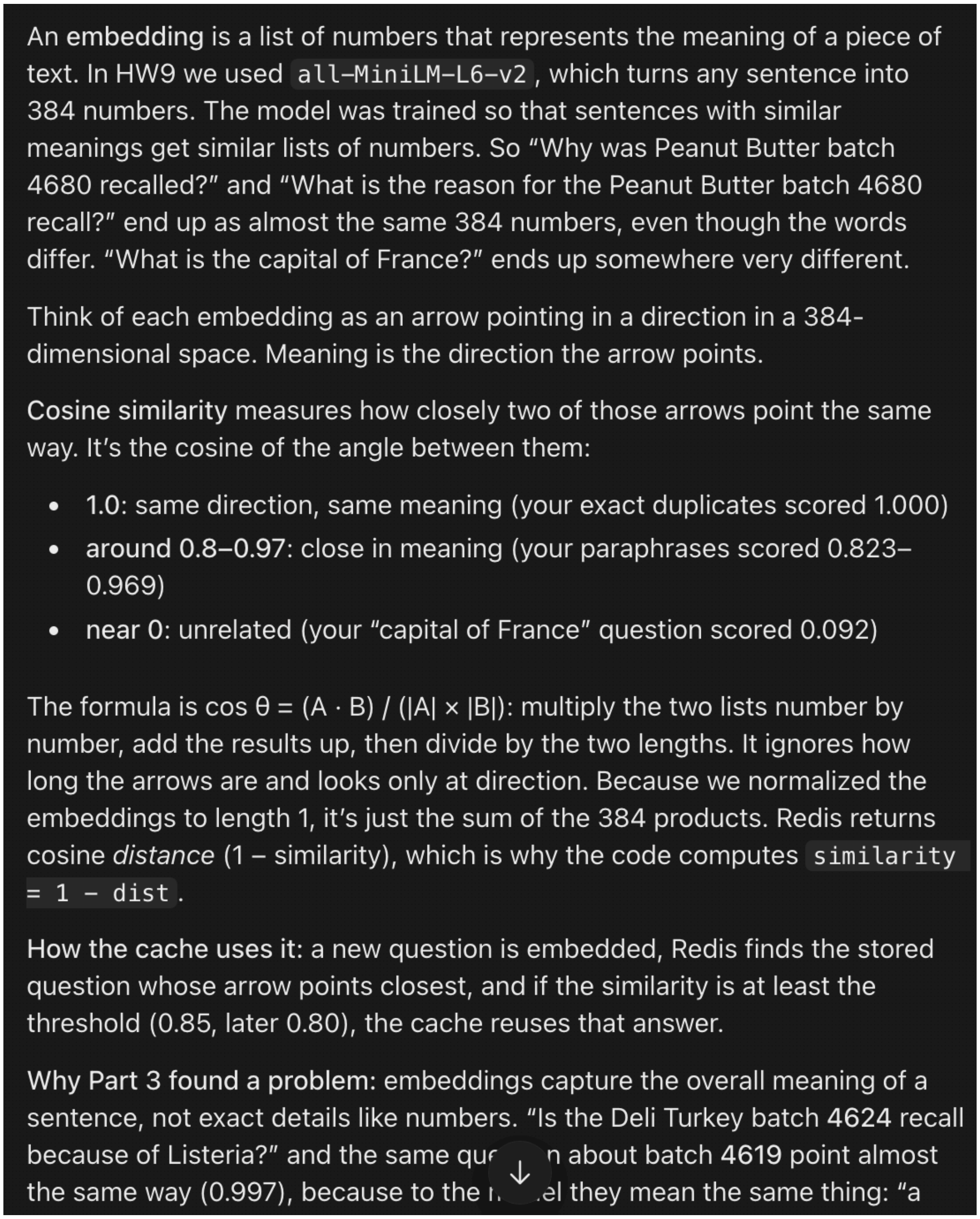
test_data.json (12 questions)

X01	How many units of Shredded Cheese batch 4610 were recalled?
X02	Which lot is affected by the Bagged Lettuce batch 4633 recall?
X03	Who manufactures the recalled Ice Cream batch 4650?
X04	Why was Hummus batch 4654 recalled?
X05	Is the Butter batch 4656 recall because of Listeria?
X06	What should I do if I bought Deli Turkey batch 4658?
X07	What is the notice code for the Frozen Spinach batch 4659 recall?
X08	Which company recalled the Yogurt batch 4675?
X09	How many units of Orange Juice batch 4596 are affected?
X10	Is the Peanut Butter batch 4699 recall because of Listeria?
X11	What wine pairs well with salmon?
X12	How do I sign up for the store loyalty program?

AI_use

- **What the AI assistant (Claude) did:** I used it to explain the concepts (batched database syncing for counters, embeddings and cosine similarity, HNSW indexes, threshold trade-offs, cache invalidation, evaluator–repair loops). It also proposed the labeled cases, and drafted this report.
- **What I did:** I ran every command on my machine (Docker Redis, MySQL, Kafka, Ollama), reviewed the outputs, and took all screenshots. I decided to move the Reviewer to qwen3:8b after reading its rejection reasons, and I froze the labels with a commit before the sweep.
- **Accuracy:** all numbers in this report come from my runs (`reports/hw09/raw/*.json`). The AI found bugs during the work, including a redis-py RESP3 parsing error, the HW8 rule-check false positive, and the embedding blindness to batch numbers. Each fix was

rerun
and



An **embedding** is a list of numbers that represents the meaning of a piece of text. In HW9 we used `all-MiniLM-L6-v2`, which turns any sentence into 384 numbers. The model was trained so that sentences with similar meanings get similar lists of numbers. So “Why was Peanut Butter batch 4680 recalled?” and “What is the reason for the Peanut Butter batch 4680 recall?” end up as almost the same 384 numbers, even though the words differ. “What is the capital of France?” ends up somewhere very different.

Think of each embedding as an arrow pointing in a direction in a 384-dimensional space. Meaning is the direction the arrow points.

Cosine similarity measures how closely two of those arrows point the same way. It’s the cosine of the angle between them:

- **1.0**: same direction, same meaning (your exact duplicates scored 1.000)
- **around 0.8–0.97**: close in meaning (your paraphrases scored 0.823–0.969)
- **near 0**: unrelated (your “capital of France” question scored 0.092)

The formula is cos θ = (A · B) / (|A| × |B|): multiply the two lists number by number, add the results up, then divide by the two lengths. It ignores how long the arrows are and looks only at direction. Because we normalized the embeddings to length 1, it’s just the sum of the 384 products. Redis returns cosine *distance* (1 – similarity), which is why the code computes `similarity = 1 - dist`.

How the cache uses it: a new question is embedded, Redis finds the stored question whose arrow points closest, and if the similarity is at least the threshold (0.85, later 0.80), the cache reuses that answer.

Why Part 3 found a problem: embeddings capture the overall meaning of a sentence, not exact details like numbers. “Is the Deli Turkey batch **4624** recall because of Listeria?” and the same que n about batch **4619** point almost the same way (0.997), because to the el they mean the same thing: “a

verified.